

ARUN SHANKAR S

DEVELOPER | SOFTWARE ENGINEERING

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PROFILE

Software Engineer with 1+ year of experience developing enterprise AI applications, Retrieval-Augmented Generation (RAG) systems, AI agents, and domain-specific LLM solutions. Experienced in Python, FastAPI, React, vector databases, embeddings, and enterprise AI PoCs for the automotive industry. Specialized in building production-ready AI systems for document ingestion, semantic retrieval, and autonomous reasoning.

SKILLS

- **Programming Languages:** Python, Java, JavaScript
- **Generative AI:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, Prompt Engineering, Semantic Search, Embeddings, Vector Databases
- **Framework & Libraries:** FastAPI, Flask, REST APIs, React, PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers, Pandas, NumPy
- **Databases:** PostgreSQL, MySQL, Qdrant, MongoDB, Neo4j
- **Cloud & Developer Tools:** Google Cloud Platform (GCP), Git, GitHub, Ollama

WORK EXPERIENCE

UST - Trivandrum | September 2025 – Present

Developer I Software Engineering

- Developing an enterprise AI Proof of Concept (PoC) for an automotive engineering client by integrating 3 data platforms (Neo4j, PostgreSQL, and Qdrant) to enable domain-specific knowledge retrieval.
- Designing domain-specific retrieval pipelines for multiple automotive engineering use cases involving technical documents and domain knowledge.
- Building end-to-end AI workflows covering document ingestion, parsing, chunking, embedding generation, vector search, reranking, and LLM-driven semantic search and question answering across enterprise knowledge bases.
- Optimizing retrieval quality using adaptive chunking strategies and configurable retrieval thresholds for domain-specific document retrieval.

INTERNSHIP

UST | Trivandrum | March 2025 – September 2025

- Worked on GCP and MySQL for data handling across 5+ modules, improving backend access efficiency.
- Performed data migration across 3 environments, ensuring 100% accuracy and 0% data loss.
- Collaborated with cross-functional engineering teams to integrate backend solutions into production systems supporting enterprise deployments.

PROJECTS

Adaptive Retrieval-Augmented Generation (RAG) Platform

- Developed an end-to-end Retrieval-Augmented Generation (RAG) platform enabling intelligent document retrieval and context-aware question answering over enterprise document collections.
- Designed and implemented configurable chunking strategies and adaptive retrieval thresholding to improve retrieval relevance for diverse document types and knowledge sources.
- Engineered scalable document ingestion, embedding generation, vector indexing, and semantic retrieval pipelines using Qdrant and Sentence Transformers.
- Built interactive features including chunk visualization, single-document and multi-document question answering, retrieval analysis, and search result inspection for evaluating retrieval quality.
- Tech Stack: Python, FastAPI, React, Qdrant, Sentence Transformers, all-MiniLM-L6-v2, Ollama
- GitHub link: [Adaptive RAG](#)

Autonomous AI Coding Agent

- Developed an autonomous AI coding agent integrating 6 core capabilities: project understanding, semantic code search, code editing, file creation, Python execution, and LLM-powered reasoning.
- Engineered a tool-based architecture with 6 integrated developer tools for autonomous software development across complete codebases.
- Designed a session-based memory system that transformed project files into reusable contextual knowledge, enabling efficient context reuse across multi-file software projects.
- Implemented a 4-stage agentic reasoning workflow (Planning → Tool Selection → Action → Observation) for context-aware, multi-step task execution.
- Enhanced scalability by enabling efficient understanding and navigation of large multi-file codebases without repeatedly processing the entire project context.
- Tech Stack: Python, Flask, Ollama (Gemma4), Large Language Models (LLMs).
- GitHub link: [Coding Assistant](#)

EDUCATION

- **Master of Computer Applications** | August 2023 – May 2025 CGPA : 9.42/10
Lourdes Matha College of Science and Technology | Kerala Technical University
- **Bachelor of Science in Computer Science** | October 2020 – April 2023 CGPA : 7.6/10
National College of Arts and Science | Kerala University

AWARDS AND CERTIFICATIONS

- **Data Science with Generative AI – Illinois Tech (USA), Entri Software Pvt. Ltd., NSDC (2026) :**
Developed proficiency in Python, SQL, Machine Learning, Deep Learning, NLP, Computer Vision, and Generative AI.
- **Secured 7th Rank in Master of Computer Applications (MCA)** for the 2023–2025 batch.